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ORIGINAL ARTICLE

Parametric and non-parametric learning models for time series: case study of vehicle sales based on exogenous variables in Brazil

Modelos de aprendizagem paramétricos e não-paramétricos para séries temporais: estudo de caso vendas de veículos no Brasil baseado em variáveis exógenas

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Abstract: Demand forecasting is a strategic necessity for vehicle manufacturers, directly influencing production schedules, operations management, and resource allocation. As markets become increasingly volatile, advanced mathematical models, particularly those leveraging machine learning techniques, have become indispensable. This study presents a comprehensive comparative analysis of parametric and non-parametric models for time series forecasting of vehicle demand in Brazil. By integrating key economic indicators, such as GDP growth, unemployment rates, and interest rates, we explore how these models can more accurately predict market demand and adapt to external economic variables. Through extensive computational simulations, we evaluate the accuracy, robustness, and practical applicability of these models. Our findings provide critical insights into the strengths and limitations of each modeling approach, offering valuable guidance for improving demand forecasting strategies. This research not only affirms the efficacy of learning models in enhancing demand planning but also lays a robust groundwork for future theoretical and practical advancements across various economic sectors.

Keywords: Demand forecasting; Time series; Prediction; Learning models; Automotive.

Resumo: A previsão de demanda é uma necessidade estratégica para os fabricantes de veículos, influenciando diretamente o planejamento de produção, a gestão de operações e a alocação de recursos. Com a crescente volatilidade dos mercados, modelos matemáticos avançados, especialmente aqueles que utilizam técnicas de aprendizado de máquina, tornaram-se indispensáveis. Este estudo apresenta uma análise comparativa abrangente entre modelos paramétricos e não-paramétricos para a previsão de demanda de veículos no Brasil. Integrando indicadores econômicos-chave, como o crescimento do PIB, taxas de desemprego e taxas de juros, exploramos como esses modelos podem prever de forma mais precisa a demanda de mercado e se adaptar a variáveis econômicas externas. Por meio de extensas simulações computacionais, avaliamos a precisão, robustez e aplicabilidade prática desses

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modelos. Nossos resultados oferecem insights críticos sobre as forças e limitações de cada abordagem de modelagem, fornecendo orientações valiosas para aprimorar as estratégias de previsão de demanda. Esta pesquisa não só confirma a eficácia dos modelos de aprendizado na melhoria do planejamento de demanda, mas também estabelece uma base robusta para futuros avanços teóricos e práticos em diversos setores econômicos.

Palavras-chave: Previsão de demanda; Séries temporais; Predição; Modelos de aprendizado; Automotivo.

1 Introduction

Demand planning is a complex and essential process critical to the success of manufacturing operations, especially in the highly competitive automotive industry. It ensures the availability of the right products at the right time while minimizing costs associated with overproduction, underproduction, and excess inventory. The effectiveness of demand planning has far-reaching implications, influencing a company's ability to meet customer expectations, optimize supply chain operations, and maintain profitability in fluctuating market conditions.

At the heart of demand planning is the challenge of aligning supply with customer demand, a challenge that is exacerbated by the volatility and unpredictability of market conditions. As Chopra & Meindl (2019) articulate, the ability to forecast demand accurately is critical to avoiding the costly pitfalls of excess inventory and stockouts. Similarly, Cachon (2012) emphasize the intricate balance required to manage inventory efficiently while ensuring product availability. This balance is particularly delicate in the automotive industry, where product life cycles are short, and consumer preferences can shift rapidly.

Silver et al. (2016) discuss the various costs associated with maintaining inventory, including maintenance, storage, and risk costs. They highlight that ineffective demand planning can result in high levels of idle inventory, which in turn increases these costs. Additionally, Nahmias (2008) provide detailed insights into the different types of costs related to inventory and offer strategies for effective inventory management to minimize these costs.

Research by Chen et al. (2007) shows that retail and wholesale sectors face substantial costs related to inventory performance, further underscoring the importance of accurate demand planning. Fisher & Raman (1996) also discuss the impact of demand uncertainty on inventory costs and present strategies to reduce these costs through accurate demand forecasting and responsive supply chain practices.

According to Ballou (2007), supply chain management is based on strategically made decisions using the information contained throughout the production chain. Supply chain management is described as the strategic and systematic coordination of traditional business functions and their tactics within a specific company, as well as across business operations within the supply chain. Its objective is to improve the long-term performance of individual companies and the supply chain as a whole. It is crucial to emphasize that supply chain management involves coordinating the flow of products through functions and companies to achieve competitive advantage and profitability for each company in the supply chain, as well as for all the members of the same chain.

In many sectors, including the automotive one, products have little differentiation between brands, and other factors have gained greater attention to maintaining customer satisfaction and loyalty. In sales, the main concern is to have the product available at dealerships when the customer wants to buy, as quick high-quality

responses help companies achieve their objectives. Sales volume has a significant impact on operational costs, making good demand management practices increasingly significant for companies.

Traditional time series models, including moving averages and exponential smoothing, have been widely used due to their simplicity and ease of implementation. However, these models often fail to capture the complex non-linear relationships in demand data, particularly when external economic variables influence the trends. The emergence of advanced learning models, which can incorporate these exogenous factors, represents a significant evolution in demand forecasting methodologies. These models are capable of not only identifying and adapting to underlying trends but also accounting for the broader economic context that impacts demand.

This study is motivated by the need to explore the comparative effectiveness of parametric and non-parametric models in forecasting vehicle demand within the Brazilian market—a market characterized by its economic volatility and competitive pressures. By integrating economic indicators into our models, we aim to provide a more accurate and nuanced understanding of how these factors influence vehicle demand. The findings from this research are anticipated to have significant implications, not only for automotive manufacturers but also for other sectors whereby demand forecasting plays a pivotal role in operational and strategic decision-making.

2 Materials and methods

This chapter provides a comprehensive overview of the processes used to collect and prepare the data for analysis. It delves into data collection techniques, outlining the sources, tools, and procedures utilized to gather relevant information. The methods of data preparation are then shown, including the steps taken to clean, organize, and preprocess the data, ensuring its suitability for the subsequent implementation of models and analysis. These methodological approaches are crucial in establishing the reliability and validity of the study findings.

The first step involves the collection and preparation of data, following the guidelines by Provost & Fawcett (2013). This phase focuses on gathering historical public data on automobile production and economic indicators in Brazil from January 2002 to December 2023. The data is cleaned and preprocessed to ensure accuracy and consistency. Proper data preparation is pivotal for the success of subsequent analysis and modeling.

The vehicles sales data (time series data) for this study are sourced from the ANFAVEA - National Association of Motor Vehicle Manufacturers (ANFAVEA, 2024) and exogenous data (economic variables) are sourced from IBGE - Brazilian Institute of Geography and Statistics in Brazil (IBGE, 2024). Both data are public and do not need authorization. The data collected includes the latest figures from January 2002 to December 2023, published in early January 2024.

Exogenous data, serving as economic variables, plays a vital role in this research by providing contextual insights that influence automobile production. These variables include key indicators, such as GDP rates (Gross domestic product), unemployment rates, basic interest rates and production index. By incorporating these economic factors, the analysis aims to capture the broader macroeconomic environment and its impact on the automotive industry. Understanding these relationships allows for more accurate modeling and forecasting, as economic conditions significantly affect production volumes and market dynamics. This comprehensive approach ensures that

the models account for external economic influences, thereby enhancing the reliability and relevance of the research findings.

In this research, the Python programming language (version 3.9) in the Jupyter Notebook environment and the libraries keras, TensorFlow, pandas, matplotlib, statsmodels, scikit-learn, statistics, sklearn, numpy, pycaret, and seaborn are used. The machine used is a MacBook Pro with a 2.4 GHz Intel Core i5 Quad-Core processor, Intel Iris Plus Graphics 655 1536 MB graphics card, 8 GB 2133 MHz LPDDR3 RAM, and operating system Sonoma 14.2.1 (23C71).

The public data used in this research includes statistical information from January 2002 to December 2023 on a monthly basis. The vehicle types are cars and small commercial vehicles. Table 1 presents the type of data collected and its statistical information.

Table 1. Statistical table.

	SALES	GDP	UNEMP_RATE	SELIC	PROD_INDEX
count	264	264	264	264	264
mean	208,341.97	429,136.71	14.12	0.92	124.43
min	1048	111,477.10	7.50	0.13	20.00
25%	173,386.75	228,375.92	11.10	0.68	105.41
50%	203,449.5	422,934.00	14.20	0.92	121.91
75%	251,242.75	567,201.08	17.00	1.12	145.50
max	330,462	940,000.00	20.70	2.08	182.70
std dev	53,261.35	223,524.54	3.44	0.38	27.45

Table 1 lists the individual statistical value for sales and economics data.

This research used the outlier treatment by Rolling Statistics (Mean) (Hyndman & Athanasopoulos, 2019), which refers to a statistical technique used in time series analysis to calculate the moving average of a dataset over a specified time window. This involves computing the mean (average) of a subset of data points within the rolling window, which is then shifted forward one observation at a time. By utilizing rolling statistics with the mean, the analysis can smooth out fluctuations and identify trends or patterns in the data over time. It is particularly useful for detecting changes in the underlying behavior of a time series while minimizing the impact of short-term variability or noise. Figure 1 shows the outliers treated.

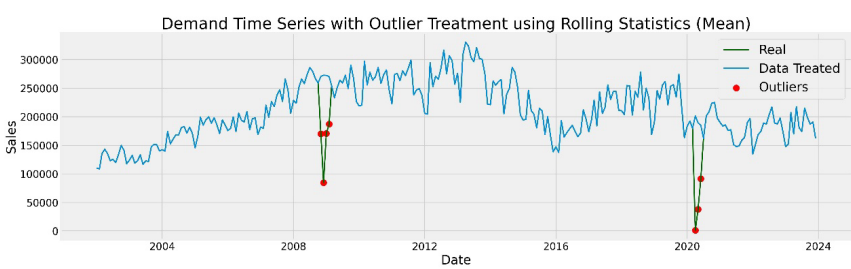

Figure 1. Sales with Outliers treated.

The division of data into training and testing sets is a critical step in predictive modeling, aimed at ensuring the developed model proper validation. In this study, 12

months of data were allocated to the testing set, a common practice to evaluate the generalization of models capability on unseen data. According to Hyndman & Athanasopoulos (2015), an appropriate division between training and testing sets is essential to avoid overfitting (Figure 2), in which the model becomes excessively tailored to the training data and fails to accurately predict new data. The choice of testing set is supported by the literature, which suggests that longer testing periods allow for a more robust assessment of the performance of models in time series forecasting scenarios (Tashman, 2000). This approach ensures that the model maintains consistent performance across different seasons and seasonal variations, providing a more realistic view of its effectiveness in practical applications.

Figure 2. Training and Test sets.

3 Literature review

Recent studies have explored the use of machine learning and advanced forecasting techniques in the automotive sector. For instance, Silva (2022) examined the impact of machine learning algorithms and big data on demand forecasting, highlighting a 37% improvement in accuracy compared to traditional qualitative methods. The study underscores the effectiveness of advanced computational techniques in demand management, particularly when incorporating causal variables into forecasting models.

Furthermore, Tamachiro et al. (2024) conducted a systematic literature review and bibliometric analysis on the application of machine learning models for time series forecasting in supply chains. The study provides an overview of trends and advancements in demand forecasting between 2011 and 2024, demonstrating the growing role of artificial intelligence in enhancing predictive performance in the automotive sector.

Another notable study by Jesus et al. (2024) explores the use of quantum neural networks for demand forecasting in vehicle financing. Their findings suggest that quantum-based models exhibit predictive capabilities comparable to classical models while requiring fewer training parameters and converging in fewer iterations. This study highlights the potential of quantum computing in the field of demand forecasting, pointing toward future developments in the integration of quantum algorithms with machine learning approaches.

These recent contributions reinforce the importance of incorporating emerging technologies such as deep learning, big data, and even quantum computing in demand forecasting models. Future research could explore hybrid approaches that integrate traditional statistical methods with AI-driven techniques, further enhancing forecasting accuracy and adaptability to market fluctuations.

A literature review on parametric and non-parametric learning models is conducted. This step involves understanding the theoretical underpinnings and practical

applications of these models in the context of vehicle production and economic data. Following the literature review, these models are implemented using the collected data. This involves setting up the models, training, and tuning hyperparameters to optimize performance.

3.1 Parametric and non-parametric models

Parametric models summarize data with a set of parameters of fixed size, independently of the number of training instances. These models often make assumptions about the underlying data distribution, such as linear relationships in linear regression. Due to their fixed number of parameters, parametric models are generally simpler and faster to train and use (James et al., 2013). This simplicity, however, can lead to underfitting, whereby the model fails to capture the complexity of the data. Examples of parametric models include linear regression, logistic regression, and polynomial regression.

Non-parametric models do not assume a fixed form for the function to be estimated, allowing the data to determine the model structure. These models have a flexible number of parameters that grow with the amount of training data, which provides greater adaptability to data complexity. Non-parametric models make fewer assumptions about the underlying data distribution, which allows them to capture more intricate patterns (Murphy, 2012). However, this flexibility comes with a higher risk of overfitting, especially with small datasets, as they may model the noise in the training data. Examples include K-Nearest Neighbors, decision trees, and support vector machines with certain kernels.

Deep learning models, particularly Long Short-Term Memory (LSTM) networks, have gained popularity for time series forecasting due to their ability to capture long-term dependencies and non-linear patterns. LSTMs are designed to handle sequential data effectively, making them a strong candidate for demand forecasting. However, despite their advantages, LSTMs require significantly larger datasets for proper training and are computationally expensive compared to traditional statistical and machine learning models. Additionally, their black-box nature makes interpretability a challenge, which is a crucial factor in business decision-making processes. Given the structured nature of our data, the relatively small dataset, and the need for model transparency, this research opted for ARIMAX, Random Forest, and XGBoost, which offer a balance between accuracy, interpretability, and computational efficiency. Future research could explore the integration of LSTMs or hybrid approaches that combine statistical models with deep learning techniques to enhance forecasting accuracy.

The crux of this study lies in the implementation of three distinct models: ARIMAX, Random Forest, and XGBoost. These models were chosen for their proven effectiveness in time series forecasting and their ability to handle both parametric and non-parametric approach.

3.1.1 ARIMA – Autoregressive Integrated Moving

The Autoregressive Integrated Moving Average (ARIMA) model is the most statistic parametric common time series model used. All models derived from ARIMA can include autoregressive models, moving averages, or mixed models. These models consist of three components. The first component is the Autoregressive model, denoted by AR(p)

(Taha Abd et al., 2021). The second component is the Moving Average model, indicated by MA(q). The third component represents the degree of integration, denoted by (d), which refers to the number of differences required to make the time series stationary. The mathematical formula for the ARIMA (p, d, q) model is (Equation 1):

$$y_t = \mu + \varphi_1 y_{t-1} + \cdots + \varphi_p y_{t-p} + e_t - \theta_1 e_{t-1} - \cdots - \theta_q e_{t-q} \quad (1)$$

where:
• $\boldsymbol{y_t} = \nabla^{\boldsymbol{d}} \boldsymbol{Y_t}$, ***and*** $\nabla^{\boldsymbol{d}}$ represent the displacement factor (The differences d for time series $\boldsymbol{Y_t}$), that is $\boldsymbol{y_t} = \nabla \boldsymbol{Y} = \boldsymbol{Y_t} - \boldsymbol{Y_{t-1}}$

The ARIMAX model is often regarded as a derivative of the ARIMA model. The key difference between them is that, in addition to using autoregressive and moving average parameters, like ARIMA, the ARIMAX model also incorporates exogenous variables (Bennett et al., 2014). Lydia et al. (2016) concluded that a significant advantage of the ARIMAX model is its ability to include variables that may be related to the dependent variable, which, in the present study, is represented by the economic indicators.

3.1.2 Random forest

Random Forest is an ensemble non-parametric machine learning method that combines multiple decision trees to improve predictive performance and robustness. It constructs a multitude of decision trees during training and outputs the class that is the mode of the classes (classification) or mean prediction (regression) of the individual trees (Breiman et al., 2017). Random Forest is highly effective for both classification and regression tasks, and it is less prone to overfitting compared to individual decision trees. It can handle large datasets with high-dimensional feature spaces and is relatively insensitive to noisy or irrelevant features.

Random Forest builds multiple decision trees during training. The number of trees *T* is a hyperparameter set by the user. Each tree is trained on a different bootstrap sample of the training data (sampling with replacement). Additionally, at each node split of each tree, a random subset of features is considered, which helps to decorrelate the trees and improve generalization. For classification tasks, the final prediction of the Random Forest is the mode of the predictions of its constituent trees. For regression tasks, it is the mean of the predictions.

The prediction of a Random Forest for an input *x* can be represented as Equation 2:

$$y_{RF}(x) = \frac{1}{T} \sum_{t=1}^{T} y_t(x) \quad (2)$$

where:
• $y_{RF}(x)$ is the predicted output of the Random Forest for input x,
• T is the total number of trees in the forest,
• $y_t(x)$ is the predicted output of the t-th tree.

Random Forests are a powerful ensemble learning method widely used in various machine learning applications due to their high accuracy and robustness. By aggregating multiple decision trees and utilizing random feature subsets, Random

Forests mitigate the limitations of individual trees and provide reliable predictions across diverse domains.

3.1.3 XGBOOST (eXtreme Gradient Boosting)

XGBOOST (eXtreme Gradient Boosting) is a non-parametric powerful and efficient implementation of the gradient boosting framework. This machine learning model is widely recognized for its speed and performance, particularly in handling large datasets and complex data structures. XGBOOST has become a popular choice among data scientists and machine learning practitioners due to its scalability, flexibility, and ability to deliver accurate predictive models (Chen & Guestrin, 2016; Zhang et al., 2018).

The core idea of XGBOOST is to build an ensemble of weak learners (typically decision trees) in a sequential manner, whereby each new tree aims to correct the errors made by the previous ones. The objective function of XGBOOST can be formulated as follows (Equation 3):

$$Obj(\theta) = \sum_{i=1}^{n} L(y_i, Y_i) + \sum_{k=1}^{k} \Omega(f_k) \tag{3}$$

where:
• L is a differentiable convex loss function that measures the difference between the predicted output *Yi*, and the true output *yi*,
• Ω is a regularization term that penalizes the complexity of the model to avoid overfitting,
• n is the number of training samples,
• K is the number of trees in the model,
• f_k represents the k-th tree in the model (Chen & Guestrin, 2016).

The regularization term Ω typically includes components such as the number of leaves in a tree and the L2 norm of the leaf weights (Equation 4):

$$\Omega(f) = \gamma\, T + \frac{1}{2}\, \lambda \sum_{j=1}^{T} w_j^2 \tag{4}$$

where:
• *T* is the number of leaves in the tree,
• w_j is the weight of the j-th leaf,
• ϒ and λ are regularization parameters (Natekin & Knoll, 2013).

XGBOOST provides a rich set of parameters to fine-tune the model performance. These parameters can be broadly categorized into three groups: general parameters, booster parameters, and learning task parameters (Chen & Guestrin, 2016; Zhang et al., 2018).

Flexibility and robustness of XGBOOST make it suitable for a wide range of predictive modeling tasks. By understanding its formula and parameters, practitioners can leverage its capabilities to build highly accurate models efficiently. Whether by careful parameter tuning or leveraging its advanced features, XGBOOST continues to be a go-to tool for machine learning competitions and real-world applications (Chen & Guestrin, 2016; Hastie et al., 2021).

Random Forest and XGBoost are both based on Decision Trees, and popular machine learning algorithms, but they differ in their approaches. Random Forest is an ensemble

method that builds multiple decision trees independently and aggregates their results, offering robustness against overfitting and noise, although it can be computationally intensive. XGBoost, in turn, is a boosting technique that builds trees sequentially, with each tree correcting the errors of the previous ones. It uses gradient-based optimization and regularization techniques, making it more efficient and often more accurate, but also more complex to tune compared to Random Forest (Chen & Guestrin, 2016).

4 Models implementation

Integrating exogenous variables with high correlation in time series analysis is crucial for improving the predictive accuracy of models. These variables, strongly related to the target variable (such as sales), provide additional insights into the factors influencing time series dynamics, leading to more robust and accurate forecasts (Maçaira et al., 2018). This leads to more robust and accurate forecasts, as the model can better account for external factors that impact the target variable. Ultimately, leveraging exogenous variables with high correlation helps improve the model ability to capture and predict complex patterns and variations in the time series data. Figure 3 shows the correlation with target variable (Sales) for training and testing sets.

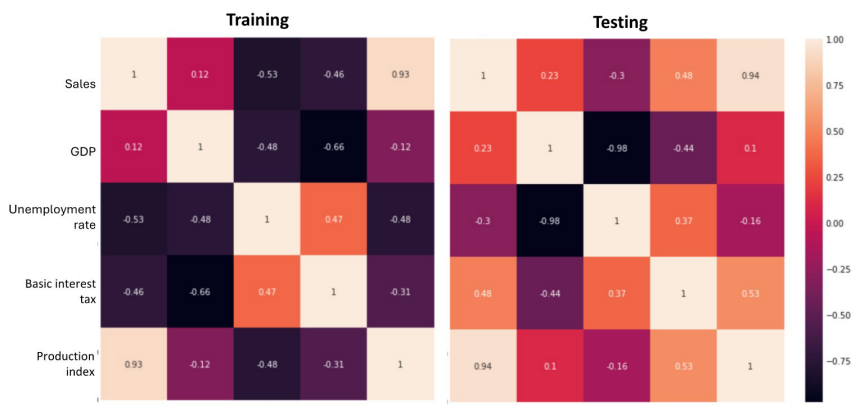

Figure 3. Correlation Matrix of Exogenous variables.

After simulations, for a robust and accurate resulting model, were selected the highest correlation relationship: Unemployment rate, Basic interest tax and Production index (Figure 4).

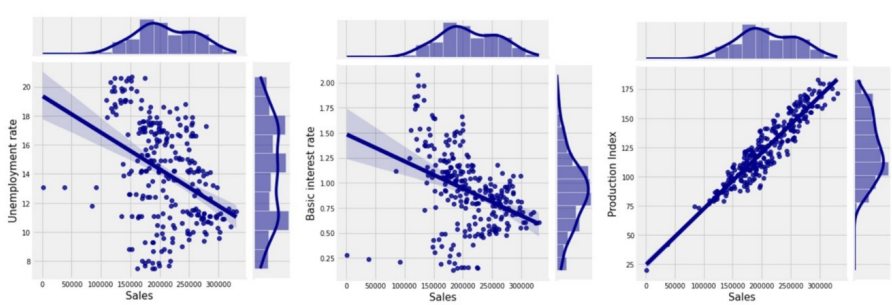

Figure 4. Sales and selected exogenous variables relationship.

4.1 Models parametrization

Parametrization techniques are used to facilitate the efficient tuning of model parameters, improving the overall performance and accuracy of the predictive models. For this study, AutoARIMA for ARIMAX and grid search for Random Forest and XGBOOST are used.

The AutoARIMA function automates the process of determining the optimal parameters for ARIMAX models. AutoARIMA systematically explores different combinations of the autoregressive (AR) parameter p, the integrated (I) parameter d, and the moving average (MA) parameter q to identify the best model based on criteria such as AIC (Akaike Information Criterion) or BIC (Bayesian Information Criterion) (Maçaira et al., 2018). By leveraging AutoARIMA, the most suitable values for these parameters are efficiently selected without the need for manual tuning. Additionally, AutoARIMA can handle the inclusion of exogenous variables, automatically determining their appropriate lag structure if they significantly influence the forecast. This approach streamlines the model selection process, ensuring that the ARIMAX model is optimally configured to capture the underlying patterns of the time series and the impact of external factors. For the context of this study, AutoARIMA selects the parameters (3, 1, 1) (1, 0, 1, 12).

For Random Forest model, Grid search is used to systematically explore a range of hyperparameters to find the optimal combination (Zhang et al., 2014). Key parameters in the grid search process include the number of trees (n trees), the number of features to consider for each split (m features), the maximum depth of the trees (max_depth), the minimum number of samples required to split a node (min_samples_split), and the minimum number of samples required at a leaf node (min_samples_leaf). Grid search evaluates the performance of the Random Forest model in a predefined grid of parameter values using cross-validation, aiming to identify the combination that yields the best performance metrics. This comprehensive search helps in fine-tuning the model, balancing the trade-off between bias and variance, and achieving optimal predictive performance. For this study, was set-up with range grid of {'max_depth': [3, 5, 7, 20, 50, 100, None], 'min_samples_split': [2, 5, 10, 50, 100], 'min_samples_leaf': [1, 2, 4], 'max_features': ['auto', 'sqrt', 'log2']}; grid search selected the parameters {'max_depth': 3, 'max_features': 'log2', 'min_samples_leaf': 1, 'min_samples_split': 5}.

Grid search function is also employed to explore various hyperparameters and identify the best settings for the XGBOOST model. Important parameters for XGBOOST include the number of boosting rounds (n rounds), which determines how many trees are built sequentially; the learning rate (η), which controls the contribution of each tree; and the maximum depth of each tree (max_depth), which affects the complexity of the model. Grid search also considers regularization parameters, such as minimum child weight (min_child_weight) and gamma (gamma), which help control overfitting. Additionally, parameters such as subsample ratio (subsample) and column sampling ratio (colsample_bytree) are tuned to ensure robustness and avoid overfitting. By systematically evaluating different combinations of these parameters using cross-validation, grid search helps find the optimal configuration for XGBOOST, thereby enhancing the model accuracy and generalization capabilities. Grid search results for XBoost, considering the context of this study and range of parameters{'n_estimators': [300, 500, 800, 1000],'learning_rate': [0.005, 0.01, 0.04],'max_depth': [7, 10, 12, 14], 'gamma': [0, 0.1, 0.2, 0.5, 1],'subsample': [0.6, 0.8, 1.0],'colsample_bytree': [0.6, 0.8, 0.9, 1.0],'reg_alpha': [0, 0.1, 0.5, 1, 10],'reg_lambda': [0, 0.1, 0.5, 1, 10]}, are {'n_estimators': [1000], 'learning_rate': [0.04], 'max_depth': [100], 'gamma': [1], 'subsample': [1], 'colsample_bytree': [1], 'reg_alpha': [0.1], 'reg_lambda': [10]}.

5 Models performance verification, findings and comparison

Once the models are implemented, their performance is verified and compared using the techniques described in Kuhn & Johnson (2013). This step includes evaluating the models based on metrics. Additionally, cross-validation techniques are employed to ensure the robustness of the models. The results are compared to identify the most effective model for predicting automobile production and understanding exogenous variables trends.

5.1 Models performance verification

To evaluate the performance of time series forecasting models, it is essential to use quantitative metrics that provide a clear view of the models accuracy and efficiency. Performance evaluation criteria MSE, MAE, MAPE, Accuracy and Process Time are used for comparing predictive modeling studies according to Shcherbakov (2013), Tayman & Swanson (1999), Hyndman & Athanasopoulos (2019) and Makridakis et al. (2020). MSE provides an absolute measure by how much the predicted results differ from the actual values (Bishop, 2006). MAE sums the absolute error values, offering a more direct representation of the total error terms (Hyndman & Athanasopoulos, 2019). MAPE is the summary measure most often used for evaluating the accuracy of population forecasts, calculates the average of the absolute percentage errors between the forecasted values and the actual values (Tayman & Swanson, 1999). Accuracy transforms the Mean Absolute Percentage Error (MAPE) into a more intuitive measure of accuracy, where higher values indicate better performance (Hyndman & Athanasopoulos, 2019).

The formulas for these criteria are (Equations 5 to 8):

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \tag{5}$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i| \tag{6}$$

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^{n} \left| \frac{y_i - \hat{y}_i}{y_i} \right| x\ 100 \tag{7}$$

$$\text{Accuracy} = 1 - MAPE \tag{8}$$

where:
n is the number of observations.
y_i is the actual value of the dependent variable for the *i*-th observation.
$\hat{y}_i$ is the predicted value of the dependent variable for the *i*-th observation.
$\overline{y}$ is the mean of the actual values of the dependent variable.

Processing time is the time required for model training and fitting, as it is a critical factor in the practical application of these models in real-world scenarios (Deb et al., 2017).

5.2 Findings and comparison

Table 2 provides a comparative analysis of three different learning models, ARIMAX, Random Forest, and XGBOOST, used to forecast vehicle demand based on

economic indicators in Brazil. The results indicate significant variations in performance, with the Random Forest model demonstrating superior accuracy and consistency compared to the Decision Tree and XGBOOST models.

Table 2. Comparative Analysis.

	ARIMA(X)	Random Forest	XGBOOST
MSE (10^9)	0.07	0.31	0.16
MAE (10^5)	0.06	0.13	0.09
MAPE %	3.92	7.91	5.68
Accuracy %	96.08	92.09	94.32
Processing Time (s)	1.19	0.61	0.05

Table 2 lists the MSE, MAE, MAPE, Accuracy and Processing time values.

The Random Forest has MAPE of 7.91%, an Accuracy around 92% and but has intermediate processing time among models, indicating robust model. The predictions have the highest deviation from the actual values, as evidenced by MSE. This indicates that the predictions have a mediate spread from the actual values as shown in the Figure 5, suggesting a good accuracy. Despite capturing some trends, the Random Forest model struggles to accurately predict the real values.

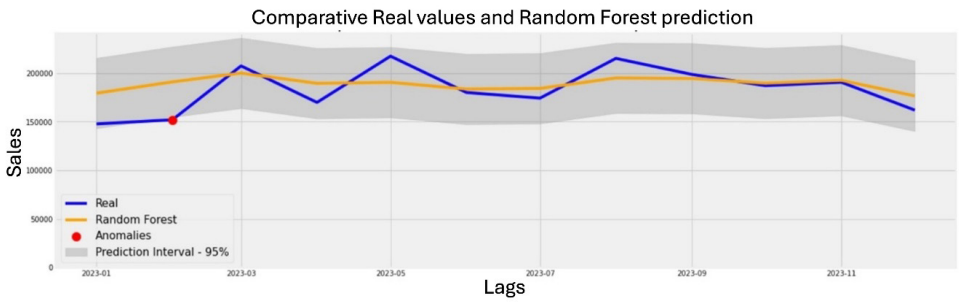

Figure 5. Comparative Real values and Random Forest prediction.

The XGBOOST model demonstrated good performance with an MAPE value of only 5.68%. Additionally, the model achieves a considerably low MSE of 0.16x(10^9), suggesting good accuracy of 94.32%. The small MAE of 0.09x(10^5), further supports the accuracy of the model, indicating, on average, small deviates from the actual values, as demonstrated in Figure 6.

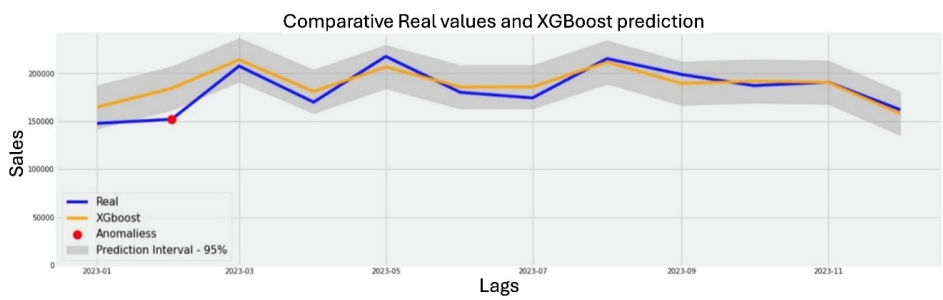

Figure 6. Comparative Real values and XGBOOST prediction.

The XGBOOST model performed better compared to the Random Forest and ARIMAX models in terms of computational processing time.

Despite the slowest processing time, the ARIMAX model stood out from Random Forest and XGBOOST, with Accuracy of 96.08% and lowest MSE, MAE and MAPE values, reinforcing its higher performance. Figure 7 shows its performance and capacity to understand times series with exogenous variables, with few anomalies in the prediction interval of 95%.

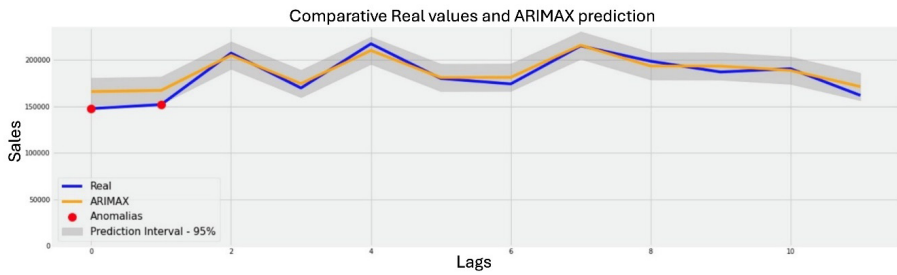

Figure 7. Comparative Real values and ARIMAX prediction.

The comparative analysis of these three models underscores the strengths and limitations of each approach. ARIMAX excels in accuracy and precision but at the cost of slower processing times, making it suitable for strategic, long-term forecasting. Random Forest offers a good balance between accuracy and speed, making it a robust choice for operational applications in which rapid predictions are necessary. XGBoost, with its combination of high accuracy and ultra-fast processing, stands out as the most versatile model, capable of handling both complex data and the demands of real-time applications.

6 Discussion

In discussing the results, guided by principles of Creswell & Creswell (2018), this study offers significant insights into the applicability of various learning models for forecasting vehicle demand in the Brazilian market. The distinct advantages and trade-offs associated with each model offer valuable guidance for selecting the appropriate forecasting tool based on specific business needs and operational contexts.

The superior performance of the ARIMAX model in terms of accuracy highlights its potential for use in scenarios where precision is critical, such as in long-term strategic planning and high-stakes decision-making. The ability of ARIMAX to incorporate external economic variables effectively makes it particularly valuable for industries heavily influenced by macroeconomic factors. For example, automotive manufacturers operating in volatile markets could leverage ARIMAX to make more informed decisions about production volumes, resource allocation, and market entry strategies.

The better performance of the ARIMAX model compared to Random Forest and XGBoost can be attributed to its ability to explicitly incorporate exogenous economic variables into the forecasting process. While machine learning models such as Random Forest and XGBoost rely on identifying patterns within the data, ARIMAX directly integrates economic indicators, allowing it to account for external market fluctuations more effectively.

In this study, key economic variables such as GDP growth, unemployment rates, and interest rates had a significant impact on vehicle demand. For example, periods of higher unemployment rates corresponded to lower vehicle sales, as consumers became more cautious with discretionary spending. Similarly, fluctuations in interest rates affected financing conditions, influencing purchasing decisions. The ARIMAX model effectively captured these relationships by leveraging the exogenous inputs, which helped it outperform machine learning models that rely solely on historical sales data.

Additionally, the relatively lower performance of Random Forest and XGBoost in this context highlights the limitations of tree-based models when handling time-dependent relationships. While these models excel in capturing complex non-linear patterns, they do not inherently consider temporal dependencies unless explicitly engineered. The results suggest that for scenarios where economic factors play a dominant role, traditional statistical models with exogenous inputs may still provide superior predictive accuracy.

Yet the Random Forest model faster processing time and solid accuracy suggest its suitability for more dynamic applications. This model could be particularly effective in environments whereby rapid response times are essential, such as in retail operations or inventory management. For instance, a retailer could use Random Forest to adjust inventory levels in real-time based on short-term demand fluctuations, thereby reducing the risk of stockouts or overstocking.

XGBoost exceptional balance between speed and accuracy positions it as a highly versatile tool, capable of being deployed in a wide range of applications. Its ability to handle large datasets quickly and accurately makes it ideal for big data environments, whereby the sheer volume of information can overwhelm less efficient models. In vehicle sales forecasting, XGBoost could be employed to optimize pricing strategies, enhance customer segmentation, and improve sales forecasting accuracy, all while maintaining the agility needed to respond to market changes in real-time.

When selecting a model for demand forecasting, it is essential to consider the specific operational requirements and constraints of the application. For long-term, high-precision forecasts, in which the accuracy of the prediction is paramount, ARIMAX is the clear choice. However, if the forecasting needs are more immediate, such as in day-to-day operations, Random Forest speed and robustness may offer the best balance. XGBoost, with its dual strengths, is particularly well-suited for applications in which both accuracy and processing speed are critical.

In the context of Vehicle Sales planning, XGBoost stands out as the best choice due to its ability to balance high forecasting accuracy with rapid processing speeds. This model is particularly advantageous in scenarios where quick responses to market changes are crucial, enabling agile adjustments to sales strategies. Furthermore, XGBoost can effectively handle large datasets and uncover complex patterns in time series, which is essential for capturing the variables that influence vehicle demand. This combination of precision and operational efficiency makes XGBoost the ideal option for optimizing planning and decision-making in this context, surpassing models that focus solely on long-term forecasting or speed.

7 Conclusion

This study presents a detailed comparative analysis of ARIMAX, Random Forest, and XGBoost models for vehicle demand forecasting in Brazil, each showcasing unique strengths and applications. ARIMAX proved to be the most accurate model, making it

ideal for strategic long-term forecasting. Meanwhile, Random Forest offered a balanced performance in terms of accuracy and speed, and XGBoost excelled in both, making it a versatile choice for diverse forecasting needs.

The insights gained from this research underscore the importance of selecting the right model based on specific business requirements and operational contexts. By understanding the trade-offs between accuracy, processing speed, and model complexity, businesses can make more informed decisions that enhance their forecasting capabilities and overall operational efficiency.

Future research should build on these findings by exploring new markets, industries, and model types, as well as incorporating real-time data and additional external variables. As the field of demand forecasting continues to evolve, the integration of more sophisticated models and data sources will undoubtedly play a critical role in driving innovation and improving decision-making processes in industries.

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Statement on Data Availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

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Alessandro Canova: Conceptualization, Methodology, Investigation, Data curation, Formal analysis, Writing – original draft, Writing – review & editing. Carlos Eduardo Cugnasca: Supervision, Writing – review & editing. All authors have read and approved the final manuscript.